Joanna Bulla

Ssr Python Developer • AI Agent & Backend Specialist

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Location: Resistencia, Chaco / Argentina

PROFESSIONAL SUMMARY

Software Engineer and Python Developer with 5+ years of hands-on experience specializing in **Agentic AI architectures**, **RAG systems**, and high-performance backend microservices. Proven track record building autonomous multi-agent pipelines with **LangGraph** and **Model Context Protocol (MCP)** at enterprise scale, as well as robust RESTful architectures with **FastAPI**, **Django**, and **.NET**. Adept at leveraging modern AI-assisted engineering workflows with **Claude** and **GitHub Copilot** to deliver reliable, production-grade solutions.

CORE TECHNICAL COMPETENCIES

GENERATIVE AI & AGENTIC SYSTEMS

LangGraph | OpenAI API | Google Gemini | CopilotKit | MCP (Model Context Protocol) | RAG Systems | DeepEval

AI-ASSISTED DEV & METHODOLOGIES

Claude (Anthropic) | GitHub Copilot | Prompt Engineering | Unit Testing (pytest) | Microservices | Agile / Jira

BACKEND & PROGRAMMING LANGUAGES

Python 3.12 / 3.11 | FastAPI | Django | Flask | C# / .NET | Java | JavaScript

DATABASES, CLOUD & DEVOPS

PostgreSQL | MongoDB | MySQL | Docker | Azure Blob & DevOps | Git

PROFESSIONAL EXPERIENCE

Enterprise & Client Engagements

Python & AI Engineer • YPF CURRENT

2025 – Present

Enterprise AI development, autonomous multi-agent systems, and specialized retrieval architectures.

- Architected and developed enterprise autonomous AI agents utilizing **LangGraph**, **Python**, **OpenAI**, and **CopilotKit**, enabling multi-agent collaboration and dynamic tool orchestration.
- Engineered, deployed, and assisted in building high-accuracy **RAG (Retrieval-Augmented Generation)** pipelines to augment LLM contextual understanding with organizational enterprise data.
- Spearheaded the creation and consumption of **MCP (Model Context Protocol)** servers and clients, establishing standardized and secure protocol layers between language models and enterprise tools.
- Accelerated SDLC velocity and elevated code quality by systematically leveraging modern AI-assisted engineering tools including **Claude** and **GitHub Copilot** across daily development.

Tech Stack: Python 3.12, LangGraph, OpenAI, CopilotKit, RAG, MCP (Model Context Protocol), Claude, GitHub Copilot, Docker, Git

Globant | **Python Developer** • *August 2021 – 2024*
Delivered enterprise software solutions and cutting-edge digital products for top-tier international clients based in the United States, including Ernst & Young (EY) and ACV Auctions.

Backend Team Lead & Developer • EY Skill Chatbot

Enterprise FinTech

Interactive conversational skill assessing creditworthiness for EY client transactions.

- Served as **Team Lead** for the development team, overseeing 3 Data Engineers, conducting code reviews, setting technical standards, and maintaining Python backend architecture.
- Engineered a modular, layered architecture from scratch with **FastAPI**, cleanly decoupling service endpoints from business domain logic.
- Constructed conversational user experiences using **Adaptive Cards**, designing dynamic back-and-forth card state flows for structured user queries.
- Structured and tuned a **PostgreSQL** database for stateful conversational tracking; containerized the service via **Docker** and orchestrated deployment on **Kubernetes**.

Tech Stack: Python, FastAPI, PostgreSQL, Kubernetes, Docker, Adaptive Cards, REST APIs, Git

AI Backend Developer • EY Capital

Generative AI & Evaluation

Enterprise RAG architecture and automated semantic response evaluation framework.

- Developed an advanced **RAG (Retrieval-Augmented Generation)** system integrated with OpenAI to dynamically retrieve relevant corporate data and generate context-aware responses.
- Implemented **DeepEval** to benchmark semantic accuracy, hallucination rates, and contextual relevancy of AI outputs, guaranteeing production reliability.
- Built an interactive test and evaluation dashboard using **Streamlit** for real-time stakeholder feedback and prompt iteration.
- Implemented high-throughput REST API endpoints in **FastAPI** with comprehensive unit testing suites in **pytest**.

Tech Stack: Python, FastAPI, OpenAI API, DeepEval, RAGAS, Streamlit, MongoDB, Docker, Git

Full-Stack / Backend Developer • EY AIHub

Internal AI Skills Platform

Internal platform allowing corporate staff to configure and commission customized AI skills.

- Implemented backend endpoints in **C# (.NET)** for requirements gathering, automated approvals, and skill lifecycle management.
- Designed secure document ingestion services leveraging **Azure Blob Storage** to allow clients to upload reference documentation.
- Integrated **SendGrid** for real-time notification dispatching to clients regarding review states and deployment proposals.
- Connected services to a centralized **PostgreSQL** database and maintained extensive unit test suites.

Tech Stack: C#, .NET, PostgreSQL, Azure Blob Storage, SendGrid, Unit Testing, Azure DevOps

PROFESSIONAL EXPERIENCE (CONTINUED)

Automotive e-Commerce

Python Backend Developer • **ACV Auctions - Capital** — **Wholesale Auto Platform (USA)**

High-volume digital marketplace for dealer auto auctions and logistics.

- Maintained and optimized auction marketplace services, building efficient multi-parameter search and filtering mechanisms in **Django**.
- Developed transactional billing modules and an end-to-end carrier route tracking system from auction win to vehicle delivery.

Tech Stack: Python, Django, MySQL, Docker, Git, Unit Testing

INDEPENDENT ENGINEERING & PERSONAL PROJECTS

AI Engineer & Creator • **Telegram Recipe AI Agent** FEATURED — **Autonomous Multimodal Agent**

Intelligent Telegram bot for multimodal recipe extraction and structured persistence.

- Engineered an autonomous AI agent in **Python** and **LangGraph** that parses recipes from photos (multimodal image extraction) and Instagram links/reels.
- Leveraged **Google Gemini** multimodal models to accurately extract ingredients, measurements, and cooking steps from unstructured media and web links.
- Designed a **FastAPI** backend handling Telegram bot webhooks, normalizing recipe data, and persisting it to a database for instant retrieval.

Tech Stack: Python, LangGraph, Google Gemini, FastAPI, Telegram Bot API, Multimodal AI, Database

Desktop App Architect & Creator • **Exam Generator App** — **GitHub**

Automated schedule generator created for a university in Resistencia, Chaco.

- Built desktop scheduling software eliminating manual administrative conflicts; developed constraint solver logic using **NumPy** and **Pandas** with a **PyQt6/PySide6** GUI.

Tech Stack: Python, NumPy, Pandas, PyQt6, PySide6, pytest, PyInstaller

Full-Stack Architect • **ChangApp Platform** — **Service Marketplace**

On-demand platform connecting local trade specialists with clients.

- Engineered REST API backend using **FastAPI** and **MongoDB**; implemented JWT authentication and **Azure Blob Storage** media uploads.

Tech Stack: Python, FastAPI, MongoDB, Azure Blob Storage, Docker, Figma, Azure DevOps

EDUCATION & CREDENTIALS

Tecnicatura Universitaria en Programación — CURRENT
Universidad Tecnológica Nacional (UTN) • Facultad Regional Resistencia, Chaco

Fullstack Developer Diploma — 2025
Universidad Tecnológica Nacional (UTN) • Buenos Aires

Idiomas / Languages — **Competencia Profesional**
Español: Nativo • **English:** Upper-Intermediate (B2/C1)